C AROUND US

C is a powerful, general-purpose programming language widely used in embedded systems, operating systems, and hardware-level programming. It is known for speed, efficiency, and direct hardware control.

MONEY AND BANKING

- ATMs use C for secure PIN and balance verification.
- Card machines process transactions in real time.
- Ensures fast, safe, and 24×7 operation.

TRAFFIC CONTROL

- Controls red, yellow, and green signals.
- Reads vehicle and pedestrian sensors.
- Prevents accidents through safety checks.

HOME APPLICATIONS

- Washing machines control wash and spin cycles.
- TVs use C for firmware and smooth performance.
- Lift systems manage floor selection and door safety.
- Alarm clocks track time and trigger alerts accurately.

ADVANCED APPLICATIONS

OPERATING SYSTEMS

- Major operating systems like Linux are largely written in C.
- Parts of Windows use C for system-level functions.
- C helps manage memory, processes, and hardware communication.
- It provides speed and direct control over system resources.

AUTOMOBILE SYSTEMS

- Engine Control Units (ECU) use C for real-time control.
- Controls fuel injection, braking, and engine timing.
- Ensures safety, speed, and reliability.
- Used in ABS and airbag systems.

WHY C ?

- ⚡ Fast and efficient
- 🔧 Close to hardware
- 🌍 Portable
- 🛡 Reliable and time-tested
- 📚 Foundation for other languages

C remains the backbone of modern computing and embedded technology.